Eutectic Solder (63Sn-37Pb) - ASTM B 32 Grade Sn63

Categories: Metal; Nonferrous Metal; Lead Alloy; Solder/Braze Alloy; Tin Alloy

Material Notes: Impurity limits in composition table are ASTM specs; other specification systems may allow other impurity limits. Solder is the largest use for tin in the U.S. Tin promotes adhesion to many base metals. This particular alloys is used as a solder in used in electronics because of its very low melting point.

Key Words: Tin Alloys; Lead Alloys

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Physical Properties	Metric	English	Comments
Density	8.40 g/cc	0.303 lb/in³	
Viscosity	1.33 cP @Temperature 280 °C	1.33 cP @Temperature 536 °F	Dynamic viscosity
	45600 cP @Temperature 280 °C	45600 cP @Temperature 536 °F	
Surface Tension	490 dynes/cm @Temperature 280 °C	490 dynes/cm @Temperature 536 °F	
	490 dynes/cm @Temperature 280 °C	490 dynes/cm @Temperature 536 °F	

Mechanical Properties	Metric	English	Comments
Hardness, Brinell	14	14	Cast
Tensile Strength, Ultimate	52.0 MPa	7540 psi	Cast solder. Typical Cu joint is 200 MPa.
Tensile Strength, Yield	43.0 MPa	6240 psi	
Elongation at Break	32 %	32 %	in 100 mm, Cast
Modulus of Elasticity	32.0 GPa	4640 ksi	Interpolated
Poissons Ratio	0.38	0.38	
Shear Modulus	12.0 GPa	1740 ksi	Calculated
Shear Strength	37.0 MPa	5370 psi	Typical soldered Cu joint is 55 MPa.
Izod Impact	20.0 J	14.8 ft-lb	Cast

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000145 ohm-cm	0.0000145 ohm-cm	

Thermal Properties	Metric	English	Comments
Heat of Fusion	37.0 J/g	15.9 BTU/lb	
CTE, linear	24.7 µm/m-°C @Temperature 15.0 - 110 °C	13.7 µin/in-°F @Temperature 59.0 - 230 °F	
Thermal Conductivity	50.9 W/m-K @Temperature 0.000 - 180 °C	353 BTU-in/hr-ft²-°F @Temperature 32.0 - 356 °F	

Melting Point	183 °C	361 °F	
Solidus	183 °C	361 °F	
Liquidus	183 °C	361 °F	

Component Elements Properties	Metric	English	Comments
Aluminum, Al	<= 0.0050 %	<= 0.0050 %	
Antimony, Sb	<= 0.50 %	<= 0.50 %	
Arsenic, As	<= 0.030 %	<= 0.030 %	
Bismuth, Bi	<= 0.25 %	<= 0.25 %	
Cadmium, Cd	<= 0.0010 %	<= 0.0010 %	
Copper, Cu	<= 0.080 %	<= 0.080 %	
Iron, Fe	<= 0.020 %	<= 0.020 %	
Lead, Pb	37 %	37 %	
Silver, Ag	<= 0.015 %	<= 0.015 %	
Tin, Sn	63 %	63 %	
Zinc, Zn	<= 0.0050 %	<= 0.0050 %	

References for this datasheet.

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